Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used Throughout the code	Yes	Proper indentation maintained throughout the Python code.
2	Proper File Structure	Files and folders are logically organized	Yes	Organized into folders such as templates, static, models, dataset, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like applicant_data, model, prediction, income_type clearly describe their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as preprocess_data(), train_model(), Predict(), predict_credit_status() are descriptive .
5	Code Comments	Inline and block comments explain logic	Yes	Important logic and preprocessing steps are documented using comments .
6	Modular Design	Code is split into reusable functions/ modules	Yes	Separate modules for preprocessing, model training, prediction, and Flask application.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code eliminated through reusable functions and modules .
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling implemented for reliable execution.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python (Pandas , NumPy)	Model Training & Prediction Pipeline	High
2	Feature Engineering Module	Python	Data Cleaning & Classification	High
3	Machine Learning Model	Scikit-learn / XGBoost	Flask Web Application & IBM Watson Deployment	High
4	Prediction Function	Python	Web Application API & User Interface	High
5	Flask Application	Flask (Python)	Local Deployment & IBM Watson Cloud Deployment	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with separate folders and modular architecture.
Readability	5	Descriptive naming conventions and consistent formatting improve readability.
Reusability	5	Components are reusable across training, deployment, and prediction workflows.
Documentation / Comments	4	Code contains sufficient comments explaining important functionality.
Overall Score	4.8	High-quality, maintainable, and scalable codebase suitable for deployment.